

Trainings ▾

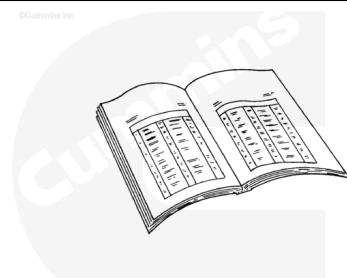
Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



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Note : Determine the appropriate radiator or coolant expansion tank cap adapter for the vehicle being serviced. A list of available caps and applications can be found in Section 14 of the Service Products Catalog, Bulletin 3377710.

Use a vacuum pump type device on the cooling system. Install the adapter on the radiator or coolant expansion tank. Connect an automotive hand held vacuum pump and apply 3 in Hg [40 in H₂O] of vacuum.

Note : Depending on the cooling system configuration, more vacuum may be required to keep the cooling system in balance.

Place a container under the aftertreatment fuel injector. There may be a small quantity of coolant that drains from the coolant lines when disconnected.

- Drain the coolant.
 - For Signature™ and ISX engines, use the following procedure. Refer to Procedure 008-018 in Section 9. ([/qs3/pubsys2/xml/en/procedures/10/10-008-018-om-auto.html](https://qs3/pubsys2/xml/en/procedures/10/10-008-018-om-auto.html))
 - For ISM, ISMe, and QSM11 engines, use the following procedure. Refer to Procedure 008-018 in

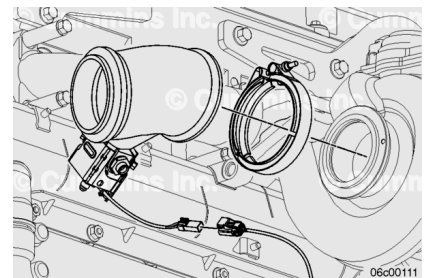
Section 9.

(/qs3/pubsys2/xml/en/procedures/35/35-008-018-om-auto.html)

- Disconnect the fuel lines and coolant lines **only** if removing the injector from the engine for cleaning or replacement. Refer to Procedure 011-051 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- Disconnect the wiring harness from the aftertreatment injector.

Remove

Remove the adaptor tube gasket.



Remove the capscrews holding the injector.

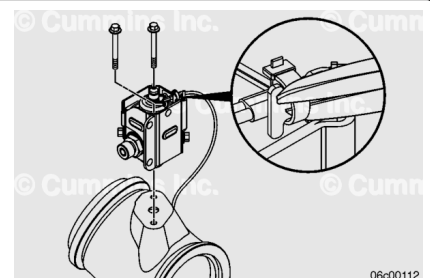
Remove the injector assembly.

Cut the EZ tie holding the injector harness to the heat shield.

Discard the metal gasket.

Discard the fibrous insulator.

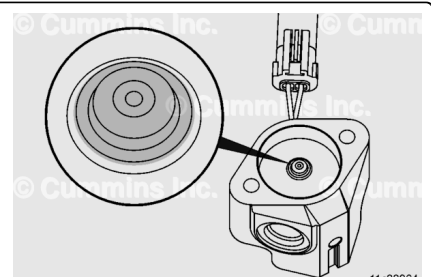
Discard the capscrews



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

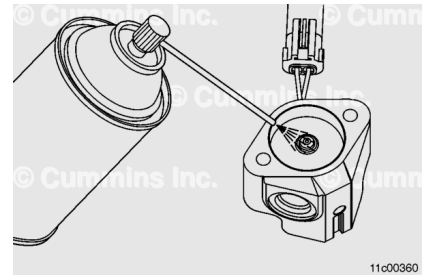
Inspect the injector tip.

Some carboning can occur on the tip of the injector.

Spray the tip of the aftertreatment fuel injector with carburetor cleaner. When spraying, focus the spray on the tip of the injector. Use enough carburetor cleaner to fill the cavity surrounding the tip and to cover the tip.

Allow the carburetor cleaner to sit for 15 seconds so that it can break down and penetrate the debris that has collected in the tip area.

Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.



⚠ CAUTION ⚠

Use only a stiff brass brush with undamaged bristles to clean the tip of the aftertreatment fuel injector.

⚠ CAUTION ⚠

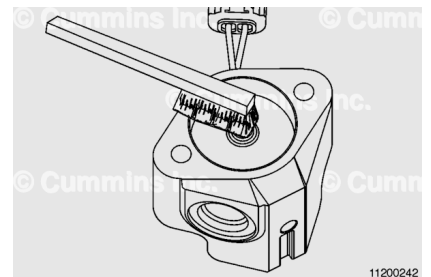
The use of a brush with damaged bristles will not be as effective.

⚠ CAUTION ⚠

The use of a steel wire brush or a steel wire wheel will cause permanent damage to the aftertreatment fuel injector.

Use a brass brush to agitate the carburetor cleaner, focusing on the tip of the injector.

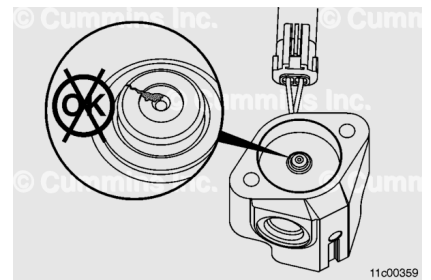
Repeat spraying and brushing for 1 minute. Often the carbon will be removed from the tip in a short period, but cleaning for a full minute will remove the carbon built up in the nozzle.



Note : The carbon this procedure is attempting to remove can often **not** be seen with the unaided eye. This carbon builds up in the nozzle tip area of the injector.

Inspect the injector tip for cracks or other damage. Replace if necessary.

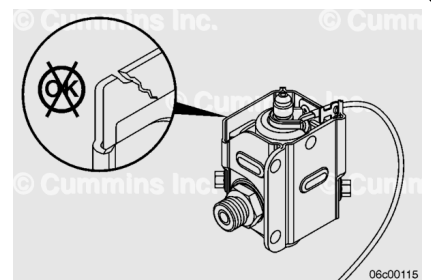
Clean the adapter tube injector mounting surface.



Inspect the heat shield, if applicable, for the following:

- Cracks
- Dents.

Replace if necessary.



Inspect for Reuse

⚠ WARNING ⚠

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- Connect the fuel line and coolant lines.
- Connect the adapter pipe to the turbocharger.
- Connect the wiring harness to the aftertreatment injector.
- If the coolant was drained, fill the cooling system.
 - For Signature™ and ISX engines, use the following procedure. Refer to Procedure 008-018 in Section 9. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-om-auto.html>)
 - For ISM, ISMe, and QSM11 engines, use the following procedure. Refer to Procedure 008-018 in Section 9. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-om-auto.html>)



- If a vacuum pump was used, remove the vacuum pump.
- Start the engine and check for leaks.

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013.html>)

Note : If a malfunction resulted in resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600. (</qs3/pubsys2/xml/en/bulletin/4021600.html>)

If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to a Cummins® Authorized Repair Location.

Install

Install a new fibrous insulator.

Install a new metal gasket.

Note : The fibrous insulator will be against the injector.

Install the aftertreatment injector onto the adapter pipe.

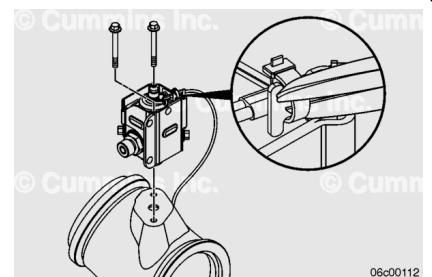
Note : Make sure the aftertreatment injector harness is **not** touching the adapter pipe.

Note : The tall side of the injector and injector heat shield should be facing the turbocharger side.

Install new capscrews. Anti-seize compound will already be applied to the capscrews.

The use of the original capscrews will result in clamp load loss and gasket leakage.

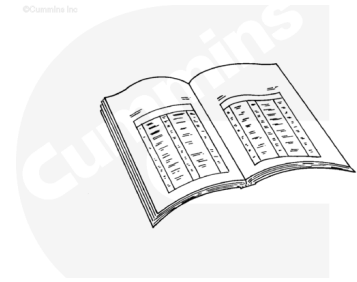
Torque Value: 11.3 n·m [100 in-lb]



Finishing Steps

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 - For ISM, ISMe, and QSM11 engines, use the following procedure. Refer to Procedure 008-018 in Section 9. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-om-auto.html>)
- If a vacuum pump was used, remove the vacuum pump.
- Start the engine and check for leaks.

If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to a Cummins® Authorized Repair Location.